St. Margaret's Co-educational English Secondary and Primary School
S.3 Information and Communication Technology
Final Exam Revision Notes

What Is Tested

	Quiz	Written Exam
Part A	Matching — Python code snippets (5 marks)	Multiple-Choice Questions (10 marks)
Part B	Short Questions — App Inventor, Python, AI, Health & Ethics (5 marks)	Matching — Python code snippets (10 marks)
Part C	—	Fill in the Blanks — ML / AI / Python / Health (10 marks)
Part D	—	Extended Questions — App Inventor, Python, AI & Future of Work (20 marks)

Section 1 — Python Basics

Data Types

Type	Keyword	Definition	Example Values
Integer	`int`	Whole numbers — positive, negative, or zero	`10, -5, 0`
Float	`float`	Numbers with a decimal point	`4.41, -0.4, 3.14`
String	`str`	Sequence of characters in single or double quotes	`'Hello world!', "London"`
Boolean	`bool`	Logical value — only True or False	`True, False`

Variable Naming Rules

- Must begin with a letter or an underscore (_) — not a number.
- Can only contain alphanumeric characters (a–z, A–Z, 0–9) and underscores.

```
score = 100         # valid — starts with a letter
_temp  = 0          # valid — starts with underscore
2player = 'Bob'     # INVALID — starts with a number
my-score = 50       # INVALID — hyphens not allowed
```

String Immutability

Strings are **immutable** — they cannot be changed after they are created.

- Trying to change a character by index causes a TypeError.

```
name = 'Hello'
name[0] = 'J'     # TypeError — strings are immutable
```

String Concatenation

Joining strings using the + operator. Only works between strings.

- Concatenating a string with an integer (non-string) causes a TypeError.

```
# Correct
print('Hello' + ' ' + 'World!')      # → Hello World!

# Incorrect — TypeError
B = 3
```

```
print('I study S' + B)              # TypeError

# Two fixes:
print('I study S', B)               # Fix 1 — use a comma
print('I study S' + str(B))         # Fix 2 — convert to str
```

Key Built-in Functions

Function	What it does	Example
`input()`	Captures user input — always returns a string	`ans = input("Enter: ")`
`int(n)`	Converts a value (string / float) to an integer	`int('42')  →  42`
`type(x)`	Returns the data type of a variable or value	`type(100)  →  <class 'int'>`
`str(n)`	Converts a value to a string — used to fix TypeError	`str(3)  →  '3'`

Common Error to Watch:

input() always returns a STRING.
A = input() → A is a string, so 3 * A repeats it, not multiplies.
Fix: A = int(input()) → now 3 * A multiplies correctly.

Code Blocks and Indentation

Python uses indentation (4 spaces) to group statements that belong together.

- Required inside if / else statements and loops.

```
score = 75
if score >= 50:
    print('Pass')    # indented — part of the if block
else:
    print('Fail')    # indented — part of the else block
```

Code Snippet Descriptions ★ (Part A Matching Reference)

Memorise these — they are the 10 snippets used in the Quiz and Written Exam matching exercises.

Ref.	Code Snippet	Description
(a)	`score = 100`	Assigns an integer value to a variable.
(b)	`price = 19.99`	Assigns a float (decimal) value to a variable.
(c)	`city = "London"`	Assigns a string (text) value to a variable.
(d)	`type(x)`	A function used to check the data type of a variable.
(e)	`print("Hello")`	Outputs a literal string or variable to the console.
(f)	`int(n)`	Converts a value (like a string or float) into an integer number.
(g)	`total = a - b`	Performs subtraction and stores the result in a variable.
(h)	`if not x == 5:`	A conditional check to see if x is not equal to 5.
(i)	`ans = input("?")`	Captures user input and assigns it to a variable as a string.
(j)	`if x <= y:`	A conditional check for less than or equal to.

Section 2 — Machine Learning for Personal Image Classification

What is Machine Learning?

Machine learning = "passing experiences to a machine" by training it.

- Objective: create a training model that can classify objects.
- The model outputs a number between 0 and 1 for each class:
  - Near 1 → high chance of being that class.
  - Near 0 → low chance of being that class.

Example Output:
Carrot Cake 0.97363 ← near 1 → classified as Carrot Cake
Siu Mai 0.02603 ← near 0 → unlikely to be Siu Mai

6 Steps to Create a Training Model ★

1. **Step 1 —** Access the Classifier Tool
   URL: https://classifier.appinventor.mit.edu/
2. **Step 2 —** Create Labels
   Define the categories the model should identify (e.g. 'Carrot Cake', 'Siu Mai').
3. **Step 3 —** Add Training Data
   Provide at least 10 images per label — by uploading or using a webcam.
4. **Step 4 —** Training
   The process of allowing the model to learn from the provided images.
5. **Step 5 —** Testing
   Provide additional images (e.g. 3 per label) to check the model's accuracy.
6. **Step 6 —** Download Trained Model
   Download the model file for use inside the App Inventor app.

Accuracy Optimization

Improving the model's reliability by modifying training data:

- Increasing the total number of images.
- Including diverse variations.
- Capturing images from different angles.
- Ensuring consistent and improved lighting.

Required App Inventor Components ★

Component	Type	Purpose
Buttons	Visible	User interaction controls.
Labels	Visible	Display classification results on screen.
WebViewer	Visible	Shows the real-time camera feed for classification.
HorizontalArrangement	Visible	Arranges components side by side.
Camera	Media	Captures photos. Found under 'Media'.
PersonalImageClassifier	Extension	AI engine — processes images, returns class name. Found under 'Extension'. File: .aix
TextToSpeech	Non-Visible	Reads classification result aloud. No visual element on screen.

Section 3 — Modules on Artificial Intelligence

AI and Computer Simulation

Two main approaches for training AI models with simulations:

Approach	How It Works
Supervised Learning (SL)	Trained on labelled data — a human tells the machine what each example is.
Reinforcement Learning (RL)	Learns by trial and error — rewarded for correct actions, penalised for wrong ones.

AI in Robotic Reasoning ★

Reasoning = the process of deciding on an action to complete a task based on available information.

Human (and AI) reasoning has THREE levels:

Level	Description	Suited For
Skill-based Reasoning	Fast, direct actions based on intuitive sensory-motor behaviour — no conscious thought.	Simple reflexes. Inefficient for complex tasks.
Rule-based Reasoning	Actions from instructions, rules, or patterns learned from prior experience.	Simple, well-defined tasks.
Knowledge-based Reasoning	Detailed analysis of the situation — integrates past experience, process knowledge, and objectives.	Complex, dynamic tasks. Primary mode for AI robots.

Key Point:
AI robots primarily use Knowledge-based Reasoning.
They learn from past experiences and adapt to uncertain conditions.

AI and the Future of Work ★

- AI can automate tasks → job replacement.
- AI can change the nature of existing jobs.
- AI can create new job opportunities.

WEF — "The Future of Jobs Report 2020": AI automation will replace and create millions of jobs by 2025.

Jobs with INCREASING Demand	Jobs with DECREASING Demand
↑ **Data Analysts and Scientists**	↓ Data Entry Clerks
↑ **AI and Machine Learning Specialists**	↓ Client Information Workers
↑ **Big Data Specialists**	↓ Customer Service Workers

OECD — "Future of Education and Skills 2030": Three skill classes humans must develop to thrive alongside AI:

Skill Class	Examples
Social and Emotional Skills	Communication, empathy, teamwork, leadership.
Cognitive and Metacognitive Skills	Critical thinking, creativity, problem-solving, learning to learn.
Practical and Physical Skills	Hands-on skills, craftsmanship, technical expertise.

Section 4 — Health and Ethical Issues

Area 1: ICT and Your Health (Physical)

Condition	Definition	Cause
Repetitive Strain Injury (RSI)	Damage to nerves and tendons.	Repeated hand/wrist movements. Example condition: Carpal Tunnel Syndrome.
Eye Strain	Fatigue and discomfort in the eyes.	Blue light from screens and fixed focus distances over long periods.
Text Neck	Pain from improper head positioning.	Tilting the head downward to look at mobile devices for extended periods.

Area 2: Equity of Access

Equity of Access = ensuring everyone has an equal opportunity to benefit from ICT, regardless of their background.

Global Digital Divide = the gap between those **with** ICT access and those **without**.

Area 3: Ethical Considerations

Protecting Privacy & Property — being careful with personal information and respecting others' digital property online.

Netiquette — professional conduct online. Three key principles:

Rule	Meaning
Be Respectful	Treat others online as you would in person. Avoid offensive language.
Critical Thinking	Evaluate information carefully before sharing or acting on it.
Security First	Protect personal information and devices; stay alert to online threats.

Quick-Reference Summary ★ Key Facts to Memorise

Python — Must Know

4 data types: int float str bool
input() always returns a STRING — use int(input()) for numbers
String + integer → TypeError — fix with str() or a comma
Strings are immutable — cannot change characters by index
Python uses indentation (4 spaces) to define code blocks

Machine Learning — Must Know

Machine learning = 'passing experiences to a machine'
Value near 1 = high chance of that class / Value near 0 = low chance
Minimum training images per label: 10
6 Steps: Access tool → Create Labels → Add Data → Train → Test → Download
WebViewer = visible / TextToSpeech = non-visible
PersonalImageClassifier.aix → found under 'Extension'
Camera → found under 'Media'

AI — Must Know

Reasoning = deciding on an action to complete a task based on available information
3 levels: Skill-based → Rule-based → Knowledge-based (most sophisticated)
AI robots primarily use: Knowledge-based Reasoning
2 simulation training methods: Supervised Learning (SL) / Reinforcement Learning (RL)
WEF 2020: AI will replace AND create millions of jobs by 2025
Jobs increasing: Data Analysts, AI/ML Specialists, Big Data Specialists
Jobs decreasing: Data Entry Clerks, Customer Service Workers
OECD 2030: Social & Emotional / Cognitive & Metacognitive / Practical & Physical

Health & Ethics — Must Know

3 physical hazards: RSI / Eye Strain / Text Neck
RSI example condition: Carpal Tunnel Syndrome
Eye Strain cause: blue light + fixed focus distances
Text Neck cause: head tilted down to look at mobile device
Equity of Access = equal opportunity to benefit from ICT
Digital Divide = gap between those with and without ICT access
Netiquette: Be Respectful / Critical Thinking / Security First

Remarks: Read all the files in the Google Classroom.